

Online Audiovisual Speaker Separation Using Efficient Visual Knowledge Distillation

Cheng Yu¹, Vahid A. Kalkhorani¹, Ashutosh Pandey², Daniel Wong², Jacob Donley²,
Buye Xu², and DeLiang Wang³

¹ Department of Computer Science and Engineering, The Ohio State University, USA

² Meta Reality Labs, USA

³ School of Data Science, The Chinese University of Hong Kong, Shenzhen, China

{yu.3500, ahmadikalkhorani.1, wang.77}@osu.edu
{apandey620, ddewong, jdonley, xub}@meta.com

Abstract

We present an efficient visual knowledge distillation (VKD) strategy to address the causal audiovisual speaker separation (AVSS) problem. Causal AVSS systems suffer from significant performance degradation due to the absence of the future context and the reliance on heavily parameterized visual embedding models originally developed non-causally. VKD progressively transfers knowledge from a causally inferred pre-trained visual front-end to a lightweight, task-oriented student model via a dynamic weighting mechanism, preserving speaker-discriminative embeddings while reducing overall system complexity. VKD compresses the visual front-end by more than 48 times in model size and 7.5 times in computational complexity. Through extensive comparisons, evaluations, and ablation studies on multiple datasets, we demonstrate that VKD combined with a top-performing online AVSS separator achieves the state-of-the-art causal AVSS performance.

Index Terms: audiovisual, speaker separation, causal separation problem, knowledge distillation

1. Introduction

In real-world acoustic environments, target speech is usually corrupted by interfering sources, such as competing speakers. The challenge of removing or attenuating interfering speech, commonly referred to as the “cocktail party problem” [1], has motivated extensive research in speaker separation (SS). Recent advances in deep learning have significantly improved SS systems, providing more effective solutions under challenging acoustic conditions [2]. Although these systems can separate mixed speech signals, they often rely on additional steps to resolve speaker permutation ambiguity [3]. To address this and other limitations, researchers have incorporated auxiliary cues into SS systems, including speaker embeddings [4] and visual embeddings [5]. In particular, visual cues are inherently robust to acoustic noise and directly associated with active speakers. Using the complementary nature of auditory and visual modalities, audiovisual speaker separation (AVSS) improves speech quality and intelligibility while naturally resolving permutation ambiguity [6].

Despite substantial advances in SS systems, most algorithms are not suitable for online deployment, as real-time applications require strict algorithmic causality. To address this limitation, several causal methods have been proposed [7–13]. However, empirical evidence indicates substantial performance gap between online and offline variants of SS systems in both audio-only (AO) and audiovisual (AV) settings [13, 14].

This gap originates from the fundamental properties of SS approaches, which rely on continuous grouping of speech features into the underlying speakers [2]. Unlike non-causal systems, which can leverage the context of a full utterance for grouping [3], causal SS systems are limited to past and present speech information. This constraint makes the grouping task more challenging and often leads to degraded separation quality [9, 13, 14]. In particular, the causal AVSS problem arises from two factors. First, the effectiveness of AV fusion is reduced when future audio and visual context are unavailable [13]. Second, most existing systems rely on pre-trained visual front-ends to enable AVSS training. However, these front-end models are typically non-causal and heavily parameterized, making them difficult to be integrated into an online framework [15]. Despite its practical relevance, research on lightweight visual front-ends remains limited [16–18].

We make two key contributions in this study. First, we extend the online AV-CrossNet [19] by applying causal inference to a pre-trained visual embedding network, DeepAVSR [20], achieving strong separation performance with full algorithmic causality. Second, we propose a simple and efficient teacher-student strategy for visual front-end module, referred to as visual knowledge distillation (VKD), which progressively transfers the representation capacity of DeepAVSR to a lightweight visual front-end. Inspired by [21], the proposed VKD is designed to learn jointly from the teacher embeddings and the AVSS task. Unlike the lightweight visual front-ends in [18], which leverage a teacher-student training with AV-HuBERT [22] and an auxiliary reconstruction objective, our approach adopts a task-oriented strategy that preserves training simplicity while achieving competitive performance relative to the teacher model. Extensive experiments demonstrate VKD’s superior results in causal AVSS on multiple datasets, while preserving computational efficiency. We also present comparisons under both non-causal and causal settings to quantify the causal separation problem, and highlight the effectiveness of our proposed approach. Ablation studies further show that our approach surpasses strong baselines and oracle frame-level permutation invariant training (tPIT) [8].

2. Methodology

In this section, we introduce the extraction process of the causal visual embeddings, the extended online AV-CrossNet for causal AVSS, and the proposed VKD strategy.

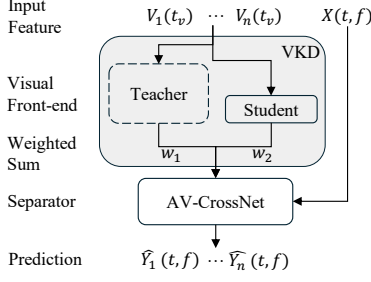


Figure 1: System diagram of the proposed approach. Dash line indicates the teacher model’s parameters are frozen.

2.1. Causal Visual Embeddings

To extract reliable speaker embeddings from visual cues, we adopt the pre-trained DeepAVSR model [20], which is widely used for its strong representation capability. However, its original architecture is non-causal and therefore unsuitable for strictly causal AVSS systems. DeepAVSR consists of a 3D convolutional front-end followed by multiple residual blocks, where the 3D convolutional kernels span both 2D image dimensions and the temporal axis. The temporal receptive field of these kernels incorporates future frames, thereby violating causality. To address this issue while retaining the pre-trained weights, we modify the inference procedure by introducing asymmetric temporal padding to the input video sequence, ensuring that each output embedding depends only on past and current frames. This adaptation preserves the speaker-discriminative representations learned during pre-training while eliminating reliance on future visual information, enabling fully causal operation without degrading embedding quality.

2.2. Online AV-CrossNet

To achieve both effectiveness and computational efficiency in AVSS, we adopt online AV-CrossNet [19] as our baseline architecture. Online AV-CrossNet separator block consists of a masked global multi-head self-attention (GMHSA) module, a cross-band module, and a narrow-band module, enabling efficient modeling of temporal and spectral dependencies under causal constraints. An optional Mamba [23] narrow-band module further enhances computational efficiency while providing strong performance. The original online AV-CrossNet is designed to address both audiovisual speech enhancement (AVSE) and audiovisual target speaker extraction (AVTSE) tasks and employs a causal visual front-end trained from scratch. However, directly applying this framework to the AVSS task leads to unstable speaker assignment and training divergence, as AVSS critically depends on discriminative and reliable speaker embeddings for consistent separation. To extend online AV-CrossNet to the AVSS setting, we replace its original visual front-end with the causal visual embedding module described in subsection 2.1. Audio and visual features are then aligned by speakers along the channel dimension, enabling parallel processing of each speaker stream while maintaining accurate correspondence. As shown in Fig. 1, the overall operation can be formulated as

$$\hat{Y}_n(t, f) = \Theta(\mathbf{X}(t, f), \mathbf{V}_n(t_v)), \quad n = 1, \dots, N \quad (1)$$

where Θ denotes online AV-CrossNet, $\mathbf{X}$ denotes the complex spectrogram of input mixture, which concatenates real and imaginary parts of short-time Fourier transform (STFT) fea-

tures. $\mathbf{V}_n$ represents the visual features, and $\hat{Y}_n$ denotes the estimated separated speech. The index n corresponds to the speaker identity among N speakers. Variables t and f indicate the time and frequency indices of the complex spectral representation, respectively, while t_v denotes the temporal index of the visual features. The network implicitly resolves speaker ambiguity by enforcing consistent alignment of each speaker’s representations along the channel dimension.

2.3. Visual Knowledge Distillation

The performance of AVSS systems heavily relies on the visual front-end, which typically involves a large number of parameters and incurs substantial computational cost. To leverage the strong representation capabilities while maintaining computational efficiency, we propose a VKD strategy. This approach progressively transfers the visual representation learning from a high-capacity DeepAVSR teacher model to a lightweight student model via a dynamic weighting mechanism during training. The teacher model comprises a 3D convolutional front-end that captures temporal visual dynamics, followed by 18 residual layers, which account for the majority of the computational overhead. To reduce complexity while preserving modeling capacity, the student model retains the same architectural structure but reduces the channel dimension of each residual layer by a factor of 16, resulting in a significantly more efficient design.

The proposed VKD is illustrated in the gray box in Fig. 1. During the initial training stage, both the pre-trained teacher and the student networks contribute to the visual stream through a weighted summation. The teacher is assigned a dominant weight w_1 , while the student is assigned a minor weight w_2 , subject to the constraint $w_1 + w_2 = 1$. As training proceeds, w_1 is linearly decreased while w_2 is correspondingly increased, progressively shifting the contribution to the student and gradually reducing the teacher’s influence. By a predefined epoch K , the teacher’s contribution becomes zero, and the visual stream is entirely handled by the student model. Then, the efficient student model is subsequently trained with the AVSS system until convergence. The student model is trained entirely on AVSS, learning to approximate the teacher’s representations while being jointly optimized for the downstream AVSS objective. Meanwhile, the teacher provides a stable optimization trajectory since the initial training phase, guiding both the student and the separator toward reliable AVSS.

3. Experimental Setup

In this section, we describe the datasets, evaluation metrics, and training hyperparameters used in our experiments.

3.1. Datasets

Following prior work in the field [6, 13], we evaluate the proposed model alongside competitive baselines on three widely used AVSS benchmarks: LRS2 [20], LRS3 [28], and VoxCeleb2 [29]. LRS2, collected from BBC broadcasts, features diverse speakers and unconstrained sentences recorded under varying illumination conditions and realistic background noise, comprising approximately 11/3/1.5 hours for training/validation/test sets. LRS3, constructed from over 400 hours of TED and TEDx talks, provides relatively clean, high-SNR recordings, with 28/3/1.5 hours for training/validation/test sets. VoxCeleb2, derived from large-scale in-the-wild YouTube videos, presents more challenging acoustic and visual conditions, including babble noise, reverberation, pose variation,

Table 1: Audiovisual speaker separation results on VoxCeleb2-2mix, LRS2-2mix, and LRS3-2mix datasets.

Method	LRS2-2mix			LRS3-2mix			VoxCeleb2-2mix		
	PESQ	SI-SDRi	SDRi	PESQ	SI-SDRi	SDRi	PESQ	SI-SDRi	SDRi
Unprocessed	1.71	0.0	0.0	1.73	0.0	0.0	1.89	0.0	0.0
Offline Systems									
AV-ConvTasNet [24]	2.69	12.5	12.8	2.58	11.2	11.7	2.17	9.2	9.8
CTCNet [25]	3.08	14.3	14.6	3.24	17.4	17.5	3.00	11.9	13.1
RTFS-Net-12 [26]	3.07	14.9	15.1	3.25	17.5	17.6	3.00	12.6	13.4
Dolphin [18]	3.29	16.8	16.9	3.36	18.8	18.9	3.17	14.6	15.1
TF-CrossNet (AO) [27]	3.28	14.4	14.7	3.38	15.6	15.9	3.26	13.2	13.6
AV-CrossNet [6]	3.56	16.8	17.1	3.67	18.3	18.5	3.41	14.6	14.9
Online Systems									
AV-ConvTasNet [24]	-	9.8	10.3	-	12.5	12.8	-	6.9	8.2
CTCNet [25]	-	8.0	8.8	-	9.4	10.2	-	2.5	4.7
RTFS-Net-12 [26]	-	12.4	12.8	-	14.0	14.4	-	8.4	10.7
Swift-Net-12 [13]	3.07	13.9	14.2	3.11	15.8	16.4	2.99	11.8	12.6
Proposed Systems									
oAV-CrossNet-Mamba	3.32	14.1	14.5	3.46	16.7	16.8	3.22	12.4	12.6
oAV-CrossNet-Mamba-VKD	3.39	14.7	15.0	3.48	16.6	16.7	3.19	11.6	11.4

and illumination changes. It contains 56/3/1.5 hours for training/validation/test sets. All speech signals are sampled at 16 kHz. The video stream is sampled at 25 FPS, and the visual input consists of grayscale region of interests (ROI) image sequences of size $t_v \times 112 \times 112$.

3.2. Evaluation Metrics

To evaluate the performance of AVSS systems, we adopt three objective metrics. The perceptual evaluation of speech quality (PESQ) [30] assesses perceptual signal quality and typically ranges from -0.5 to 4.5. In addition, scale-invariant signal-to-distortion ratio improvement (SI-SDRi) and signal-to-distortion ratio improvement (SDRi) measure the improvements in SI-SDR and SDR, respectively, relative to the unprocessed mixture. These two metrics reflect signal fidelity, with higher values indicating better separation performance. To evaluate computational complexity, we report the number of multiply-accumulate operations (MACs). A lower MACs count indicates higher computational efficiency.

3.3. Hyperparameters

To train the AVSS system, we use a combination of SI-SDR and spectral magnitude losses, where SI-SDR improves separation performance and the spectral magnitude loss improves overall signal fidelity. Training is conducted on 4 NVIDIA H100 GPUs, with each epoch comprising 5,000 mini-batches per GPU. Each batch contains 6 mixtures, where each mixture includes 3 seconds of audio and 75 frames of visual input from 2 competing speakers. For stable knowledge distillation, we set the predefined VKD epoch to $K = 50$, with initial teacher and student weights of 99% and 1%, respectively, as the model typically converges within approximately 100 epochs. Each proposed system is trained using the Adam optimizer with an initial learning rate of 0.001. We employ the PyTorch ReduceLROnPlateau scheduler to adapt the learning rate, configured with a patience of three epochs and a decay factor of 0.9. The system is considered to have converged when no improvement in the loss is observed for 20 consecutive epochs.

4. Experimental Results

In this section, we present the AVSS results on three major datasets, quantify the impact of causality on separation performance, investigate various efficient visual front-end models, and evaluate VKD against competitive baselines.

4.1. Audiovisual Speaker Separation

Table 1 reports the performance of representative AVSS systems on VoxCeleb2-2mix, LRS2-2mix, and LRS3-2mix datasets. For fair comparison, we include competitive non-causal and causal AVSS methods, along with a strong non-causal AO baseline TF-CrossNet. Among offline models, AV-CrossNet achieves the highest PESQ across all datasets and delivers the best or comparable SI-SDRi and SDRi to Dolphin, demonstrating its advantage as a baseline system.

Under causality constraints, AV-ConvTasNet and RTFS-Net-12 exhibit a significant performance degradation compared to their offline counterparts. CTCNet exhibits a more substantial gap, with SI-SDRi degradation over 6 dB, quantifying the causal AVSS problem. Swift-Net-12 narrows the gap from RTFS-Net-12 through a more efficient fusion encoder, while utilizing a full-sized visual front-end. Our proposed systems consistently outperform all existing online baselines while effectively closing the gap with offline counterparts. On LRS2-2mix and LRS3-2mix, online AV-CrossNet-Mamba-VKD (oAV-CrossNet-Mamba-VKD) surpasses all other baselines by over 0.3 in PESQ, 0.8 dB in SI-SDRi, and 0.3 dB in SDRi. On VoxCeleb2-2mix, oAV-CrossNet-Mamba-VKD outperforms the most competitive baseline, Swift-Net-12, by 0.2 in PESQ, while maintaining competitive SI-SDRi and SDRi results. Notably, oAV-CrossNet-Mamba-VKD maintains comparable performance to the uncompressed oAV-CrossNet-Mamba on VoxCeleb2-2mix and LRS3-2mix, and improves results on LRS2-2mix. Overall, oAV-CrossNet-Mamba-VKD achieves a strong balance between strict causality, separation quality, and computational efficiency, establishing state-of-the-art performance among online AVSS systems while remaining competitive with non-causal approaches.

4.2. Causal Audio-only and Audiovisual speaker separation

Table 2 presents a comparison of AO and AV systems under both online and offline settings on the VoxCeleb2-2mix dataset. The oTF-CrossNet-upit exhibits a substantial performance gap to TF-CrossNet-upit (0.6 in PESQ, and over 4 dBs in SI-SDRi and SDRi), highlighting the causal separation problem in AO systems. The oTF-CrossNet-tpit effectively narrows the gap by alleviating speaker ambiguity through oracle frame-level assignment [8]. However, this assignment is not available during inference. In contrast, the oAV-CrossNet and its Mamba variant retain strong performance with a moderate gap (around 0.2 in PESQ, and 2 dBs in SI-SDRi and SDRi). When both models employ frame-level guidance, oAV-CrossNet achieves a 0.18 improvement in PESQ over oTF-CrossNet-tpit, indicating that visual embeddings contribute additional contextual information that enhances perceptual quality. VKD further improves the performance, achieving high separation quality and substantially improving computational efficiency.

Table 2: Comparison of online and offline AO and AVSS systems on the VoxCeleb2-2mix dataset.

Method	PESQ	SI-SDRi	SDRi
Unprocessed	1.89	0.0	0.0
Audio-only Systems			
TF-CrossNet-upit	3.26	13.2	13.6
oTF-CrossNet-upit	2.67	8.5	8.5
oTF-CrossNet-tpit (oracle)	3.04	12.2	12.5
Audiovisual Systems			
AV-CrossNet	3.41	14.6	14.9
oAV-CrossNet	3.21	12.6	12.9
oAV-CrossNet-Mamba	3.22	12.4	12.6
oAV-CrossNet-Mamba-VKD	3.19	11.6	11.4

4.3. Online Visual Front-end

We have investigated several online visual front-ends on the VoxCeleb2-2mix dataset. The DeepAVSR-mid extracts features from the first residual block of the baseline front-end, whereas the one-frame approach relies on a single visual feature per speaker for the entire utterance. Although DeepAVSR-mid has the minimal size and nearly three times fewer MACs than the baseline, it performs worst, suggesting early-layer features carry limited speaker identity information. The one-frame approach achieves minimal MACs with reasonable separation quality but does not reduce parameter count. In contrast, the proposed VKD balances model size and computational cost while maintaining performance comparable to the full baseline, validating its effectiveness. We additionally evaluated two strategies: training the student front-end model from scratch with the AVSS objective and fine-tuning the knowledge-distilled student model for AVSS. In both cases, the systems did not converge and produced poor separation. Consequently, the results are omitted from Table 2. These findings indicate that naive training or direct teacher–student training alone may be insufficient, and that task-oriented, progressive knowledge transfer through VKD is essential for effective causal AVSS.

Table 3: Different visual front-end models for online AVSS on VoxCeleb2-2mix dataset.

Methods	SI-SDRi	PESQ	Params(MB)	MACs (G/s)
DeepAVSR	12.4	3.22	11.19	12.9
DeepAVSR-mid	7.1	2.65	0.17	4.4
one-frame	10.3	3.05	11.19	0.3
VKD	11.6	3.19	0.23	1.7

4.4. Visual Front-end Efficiency

Table 4 compares the visual front-ends in different AVSS systems. To our knowledge, several AVSS models perform visual front-end compression, but none of them are strictly causal. Therefore, we compare with Dolphin, the latest top-performing non-causal AVSS system that compresses the visual front-end. Similar to VKD, Dolphin employs DP-LipCoder [18], which is trained with a teacher–student framework. It distills knowledge from AV-HuBERT [22] via semantic supervision and an auxiliary image reconstruction objective. As a causal reference, Swift-Net-12 adopts the CTCNet-Lip [25] front-end and achieves 13.9 dB in SI-SDRi and 3.07 in PESQ. Compared to the CTCNet-Lip, DP-LipCoder achieves compression rates of 14.3 times in parameters and 3.3 times in MACs, with a minor performance loss (0.2 dB in SI-SDRi and 0.01 in PESQ). In contrast, VKD directly distills knowledge from the DeepAVSR while jointly optimizing the AVSS objective. It achieves much higher compression rates, with 48.6 times in parameters and 7.5 times in MACs. Meanwhile, it improves separation performance by 0.6 dB in SI-SDRi and 0.07 in PESQ. These gains confirmed that VKD enables the lightweight visual front-end to inherit the teacher’s representation capacity while being explicitly optimized for the AVSS task.

Table 4: Relative performance, size, and complexity of visual front-ends on LRS2-2mix dataset.

Methods	SI-SDRi	PESQ	Params(MB)	MACs (G/s)
Dolphin [18]				
CTCNet-Lip	17.0	3.30	11.19	7.95
DP-LipCoder	16.8	3.29	0.78	2.38
Swift-Net-12 [13]				
CTCNet-Lip	13.9	3.07	11.19	7.95
Proposed				
DeepAVSR	14.1	3.32	11.19	12.92
VKD	14.7	3.39	0.23	1.71

5. Conclusion

In this work, we address the causal audiovisual speaker separation problem by proposing visual knowledge distillation, a simple yet effective strategy for building a lightweight and fully causal visual front-end model. Through progressively transferring knowledge from a causally inferred pre-trained model to a compact student network, VKD preserves speaker-discriminative embeddings while substantially reducing model complexity. Integrated with a strong online separator, the proposed system achieves the state-of-the-art performance for online AVSS, significantly narrows the gap from non-causal models, and delivers substantial compression in both parameter count and computational cost.

6. Acknowledgments

This work was supported in part by a Meta contract and a National Science Foundation grant (ECCS-2125074) to Ohio State University, the Ohio Supercomputer Center, the NCSA Delta Supercomputer Center (OCI 2005572), and the Pittsburgh Supercomputer Center (NSF ACI-1928147). All experiments were conducted at The Ohio State University.

7. Use of Generative AI Disclosure

The author used generative AI tools to revise and refine the language in Section 1.

8. References

- [1] E. C. Cherry, “Some experiments on the recognition of speech, with one and with two ears,” *Journal of the Acoustical Society of America*, vol. 25, pp. 975–979, 1953.
- [2] D. L. Wang and J. Chen, “Supervised speech separation based on deep learning: An overview,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 26, pp. 1702–1726, 2018.
- [3] D. Yu, M. Kolbæk, Z.-H. Tan, and J. Jensen, “Permutation invariant training of deep models for speaker-independent multi-talker speech separation,” in *Proc. ICASSP*, 2017, pp. 241–245.
- [4] N. Zeghidour and D. Grangier, “Wavesplit: End-to-end speech separation by speaker clustering,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 2840–2849, 2021.
- [5] D. Michelsanti, Z.-H. Tan, S.-X. Zhang, Y. Xu, M. Yu, D. Yu, and J. Jensen, “An overview of deep-learning-based audio-visual speech enhancement and separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 1368–1396, 2021.
- [6] V. A. Kalkhorani, C. Yu, A. Kumar, K. Tan, B. Xu, and D. L. Wang, “AV-CrossNet: An audiovisual complex spectral mapping network for speech separation by leveraging narrow-and cross-band modeling,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 19, pp. 685–694, 2025.
- [7] Y. Luo and N. Mesgarani, “TasNet: Time-domain audio separation network for real-time, single-channel speech separation,” in *Proc. ICASSP*, 2018, pp. 696–700.
- [8] Y. Liu and D. L. Wang, “Causal deep CASA for monaural talker-independent speaker separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2109–2118, 2020.
- [9] H. Wang, Y. Song, Z.-X. Li, I. McLoughlin, and L.-R. Dai, “An online speaker-aware speech separation approach based on time-domain representation,” in *Proc. ICASSP*, 2020, pp. 6379–6383.
- [10] K. Li and Y. Luo, “On the design and training strategies for rnn-based online neural speech separation systems,” in *Proc. ICASSP*, 2023, pp. 1–5.
- [11] Z. Pan, G. Wichern, F. G. Germain, K. Saijo, and J. Le Roux, “PARIS: Pseudo-autoregressive siamese training for online speech separation,” in *Proc. INTERSPEECH*, 2024, pp. 582–586.
- [12] T. Ma, S. Yin, L.-C. Yang, and S. Zhang, “Real-time audio-visual speech enhancement using pre-trained visual representations,” in *Proc. INTERSPEECH*, 2025, pp. 61–65.
- [13] W. Sang, K. Li, R. Yang, J. Huang, and X. Hu, “A fast and lightweight model for causal audio-visual speech separation,” in *Proc. ECAI*, 2025, pp. 1–8.
- [14] W. Wang, Z. Pan, X. Li, S. Wang, and H. Li, “Causal self-supervised pretrained frontend with predictive patterns for speech separation,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 311–323, 2025.
- [15] P. Zhang, J. Xu, Y. Hao, and B. Xu, “Online audio-visual speech separation with generative adversarial training,” in *Proc. ICCAI*, ACM, 2021, pp. 379–385.
- [16] H. Martel, J. Richter, K. Li, X. Hu, and T. Gerkmann, “Audio-visual speech separation in noisy environments with a lightweight iterative model,” in *Proc. INTERSPEECH*, 2023, pp. 1673–1677.
- [17] C. Fan, W. Xiang, J. Tao, J. Yi, and Z. Lv, “Cross-modal knowledge distillation with multi-stage adaptive feature fusion for speech separation,” *IEEE Transactions on Audio, Speech and Language Processing*, vol. 33, pp. 935–948, 2025.
- [18] K. Li, K. Gao, and X. Hu, “Efficient audio-visual speech separation with discrete lip semantics and multi-scale global-local attention,” in *Proc. ICLR*, 2026, pp. 1–26.
- [19] C. Yu, V. A. Kalkhorani, B. Xu, and D. L. Wang, “Online AV-CrossNet: A causal and efficient audiovisual system for speech enhancement and target speaker extraction,” in *Proc. INTERSPEECH*, 2025, pp. 2985–2989.
- [20] T. Afouras, J. S. Chung, A. Senior, O. Vinyals, and A. Zisserman, “Deep audio-visual speech recognition,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, pp. 8717–8727, 2018.
- [21] K. Clark, M.-T. Luong, U. Khandelwal, C. D. Manning, and Q. Le, “Bam! born-again multi-task networks for natural language understanding,” in *Proc. ACL*, 2019, pp. 5931–5937.
- [22] B. Shi, A. Mohamed, and W.-N. Hsu, “Learning lip-based audio-visual speaker embeddings with AV-HuBERT,” in *Proc. INTERSPEECH*, 2022, pp. 4785–4789.
- [23] A. Gu and T. Dao, “Mamba: Linear-time sequence modeling with selective state spaces,” in *Proc. COLM*, 2023, pp. 1–32.
- [24] J. Wu, Y. Xu, S.-X. Zhang, L.-W. Chen, M. Yu, L. Xie, and D. Yu, “Time domain audio visual speech separation,” in *Proc. ASRU*, 2019, pp. 667–673.
- [25] K. Li, F. Xie, H. Chen, K. Yuan, and X. Hu, “An audio-visual speech separation model inspired by cortico-thalamo-cortical circuits,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, pp. 6637–6651, 2024.
- [26] S. Pegg, K. Li, and X. Hu, “Rtfs-net: Recurrent time-frequency modelling for efficient audio-visual speech separation,” in *Proc. ICLR*, 2024, pp. 1–19.
- [27] V. A. Kalkhorani and D. L. Wang, “TF-CrossNet: Leveraging global, cross-band, narrow-band, and positional encoding for single- and multi-channel speaker separation,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 4999–5009, 2024.
- [28] T. Afouras, J. S. Chung, and A. Zisserman, “LRS3-TED: A large-scale dataset for visual speech recognition,” *arXiv:1809.00496*, 2018.
- [29] J. S. Chung, A. Nagrani, and A. Zisserman, “VoxCeleb2: Deep Speaker Recognition,” in *Proc. INTERSPEECH*, 2018, pp. 1086–1090.
- [30] A. W. Rix, J. G. Beerends, M. P. Hollier, and A. P. Hekstra, “Perceptual evaluation of speech quality (PESQ)-a new method for speech quality assessment of telephone networks and codecs,” in *Proc. ICASSP*, 2001, pp. 749–752.